

Variable Coupling Matrix — Pairwise Interaction Effects

	Altitude h	Speed v	Scan angle θ	Overlap α	NFZ margin d_{nfz}
Altitude h	h <i>Detection ceiling</i>	Strong Coverage rate + detection prob.	Weak Scan efficiency	Medium Lane spacing	Medium Footprint intrusion
Speed v	Strong Coverage rate + detection prob.	v <i>Frame count</i>	Weak Independent	Weak Independent	Medium Reaction distance
Scan angle θ	Weak Scan efficiency	Weak Independent	θ <i>Turn count</i>	Weak Independent	Medium NFZ proximity
Overlap α	Medium Lane spacing	Weak Independent	Weak Independent	α <i>Redundancy</i>	Weak Independent
NFZ margin d_{nfz}	Medium Footprint intrusion	Medium Reaction distance	Medium NFZ proximity	Weak Independent	d_{nfz} <i>Safety margin</i>
Strong coupling Medium coupling Weak / independent Self (primary metric)					